How to Read Training Logs in Reinforcement Learning?

When training a reinforcement learning (RL) agent, **logs are the only reliable way to know whether learning is actually happening**. Visual behavior can be misleading. This document explains the **main indicators in the training log**, how to interpret them, and what **healthy training** looks like.

The examples refer to PPO-style logs (e.g., from Stable-Baselines3), but the principles apply broadly.

The Most Important Question

Before looking at any numbers, always ask:

> **Is the agent getting better at solving the task?**

All metrics exist to help answer this question. No single metric is enough on its own.

1. Rollout Metrics (What the Agent Experiences)

These metrics describe what happens when the agent interacts with the environment.

1.1. `rollout/ep_rew_mean` — Mean Episode Reward (MOST IMPORTANT)

What it is:
The average total reward per episode.

How to interpret it:

- Increasing over time → learning is happening
- Flat → no learning
- Decreasing → learning is unstable or reward is misdesigned

A higher reward does **not always mean the task is solved**. The agent may learn to exploit reward shaping without completing the objective.

Healthy behavior:

- Gradual upward trend
- Eventually stabilizes at a higher value than at the start

1.2. `rollout/ep_len_mean` — Mean Episode Length

What it is:
Average number of steps per episode.

How to interpret it (task-dependent):

- If the task ends when the goal is reached:
    - **Decreasing length** → agent finishes faster (good)
    - **Always near max_steps** → agent rarely finishes (bad)

- If episodes always run to max length:
    - This metric is less informative

Healthy behavior (goal-reaching tasks):

- Episode length decreases as learning progresses

2. Training Metrics

These metrics describe what happens inside the learning algorithm.

2.1 `train/explained_variance` — Value Function Quality

What it is:
How well the critic (value network) predicts future returns.

Range and meaning:

Value	Interpretation
< 0	Critic is worse than guessing
~0	Critic not learning
0.3–0.6	Learning, but still rough
> 0.7	Good understanding
> 0.9	Near convergence

Healthy behavior:

- Starts near 0
- Gradually increases
- Stabilizes above ~0.6

If this stays near zero, **policy learning will fail**, regardless of other metrics.

2.2 `train/entropy_loss` — Exploration Level

What it is:
Measures how random the policy's actions are.

How to interpret it:

- High entropy → random exploration
- Low entropy → deterministic behavior

Healthy behavior: Starts high (random policy) > Gradually decreases > Stabilizes at a moderate value

Red flags:

- Drops to near zero very early > premature convergence
- Never decreases > agent is confused or reward signal is weak

2.3 `train/std` — Action Distribution Spread

What it is:
Standard deviation of the policy's action distribution (for continuous actions).

Healthy behavior:

- Starts large
- Decreases as policy becomes confident
- Does not collapse to zero too early

If `std` is very small early in training, exploration has effectively stopped.

2.4 `train/approx_kl` and `clip_fraction` — Policy Update Size

These indicate how aggressively PPO updates the policy.

Healthy ranges (rough guidelines):

- `approx_kl`: ~0.005–0.02
- `clip_fraction`: ~0.05–0.2

Values far outside these ranges may indicate instability, but **these metrics are secondary**. They matter only if learning is unstable.

Metrics That Are Often Misinterpreted

`train/loss`, `train/value_loss`

- Absolute values are not meaningful
- Useful only for detecting explosions or NaNs
- Do not judge learning success by these alone

What a Healthy Training Run Looks Like (Summary)

A healthy PPO training run typically shows:

1. **Episode reward increases**
2. **Episode length decreases** (for goal-reaching tasks)
3. **Explained variance rises above ~0.6**
4. **Entropy decreases gradually**
5. **Policy becomes smoother and more consistent**
6. **Deterministic evaluation improves over time**

All of these should be consistent with **observed behavior**.

Common Failure Patterns and Their Meaning

Symptom	Likely Cause
High reward, no task completion	Reward hacking
Flat reward, flat variance	No learning signal
High variance, unstable reward	Learning rate too high
Early entropy collapse	Exploration too weak
Long episodes forever	Terminal reward too weak

Final Rule

> **Never trust a single metric.**
> **Learning is confirmed only when logs, evaluation, and behavior all agree.**

Understanding logs is not optional in reinforcement learning — it is a core skill.